

# NER Landslide Early Warning System - AI/ML Model Cards

Engineering Transparency & Verified Benchmark Evaluation

## 1. Static Landslide Susceptibility Index (LSI) - XGBoost

- Model Type: Gradient Boosted Decision Tree (XGBClassifier) in Scikit-Learn Pipeline
- Module Path: ml-engine/susceptibility/susceptibility\_model.py | Artifact: baseline\_xgb\_model.joblib
- Training Data: 800 georeferenced synthetic samples calibrated across 8 NER morphometric zones (ml-engine/data/synthetic\_landslide\_ner\_dataset.csv created via generate\_dataset.py).
- Split: 80% Train (640) / 20% Test (160 samples), stratified.

### Verified Test Set Evaluation Metrics (160 Unseen Samples):

| Evaluation Metric    | Random Forest Baseline | XGBoost Champion | Winner / Note                            |
|----------------------|------------------------|------------------|------------------------------------------|
| Accuracy             | 68.75%                 | 69.37%           | XGBoost (+0.62%)                         |
| Precision            | 70.24%                 | 71.08%           | XGBoost (+0.84%) - Lower false alarms    |
| Recall (Sensitivity) | 70.24%                 | 70.24%           | Tied - Captures 70.2% slope hazards      |
| F1-Score             | 70.24%                 | 70.66%           | XGBoost Champion                         |
| ROC-AUC              | 77.24%                 | 77.88%           | XGBoost (+0.64%) strong class separation |

### Key Feature Importances (Tree Gain):

1. Dense Forest Root Cohesion: 19.15% (Protective) | 2. Barren Land / Quarry: 12.46% (Hazard)  
3. Antecedent 48h Rain: 12.33% | 4. Soil Saturation: 11.50% | 5. Slope Angle: 9.03% | 6. Road Cut: 8.87%

## 2. Dynamic 2-6h Nowcast (PyTorch LSTM)

- Architecture: 2-Layer PyTorch LSTM (LandslideLSTM) in ml-engine/nowcast/lstm\_nowcast.py
- Input: 6-hour sequential vectors [hourly\_rainfall\_mm, soil\_saturation\_pct].
- Training Status: ARCHITECTURE FULLY IMPLEMENTED; NOT YET TRAINED ON REAL TEMPORAL SEQUENCES.
- Metrics: NOT YET EVALUATED (Truthful reporting: high-frequency time-series datasets for NER slopes are unavailable in open access; forward pass operates with calibrated physics surge gates).

## 3. Medium-Term 24-48h Forecast (Deterministic Load Index)

- Implementation: ml-engine/nowcast/forecast\_model.py (WeatherForecastRiskModel)
- Formula:  $load\_index = 0.5 * (rain\_proj / 250) + 0.3 * (soil\_sat / 100) + 0.2 * (slope / 60)$
- Status: Fully operational forward evaluation engine powering frontend Recharts time-series graph.

## 4. Road Disruption Graph (NetworkX)

- Implementation: ml-engine/spatiotemporal/road\_network\_graph.py (RoadNetworkRiskPropagator)
- Algorithm: Deterministic Graph Reachability (nx.has\_path & shortest\_path); NOT a GNN.
- Graph: 13 arterial nodal transit hubs, 11 key highway segments (NH-10, NH-6, SH-5, NH-29).